Improving Fabric Defect Classification via Automatic Mining and Leveraging of Category Hierarchy

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***Abstract*—Fabric defect classification is crucial in textile quality inspection, enabling category-specific defect handling. Although dozens of defect categories have been defined, annotation granularity in practice often ranges from binary (defect versus nondefect) to fine-grained multiclass classification. Typically, the annotated defect categories can be structured hierarchically or further subdivided based on domain knowledge or visual patterns. In this study, we propose a method to improve the deep classifier for the annotated categories by mining subcategories and leveraging constructed category hierarchies. Based on the annotated categories, we introduce two primary strategies: 1) mining subcategories to generate fine-grained pseudo-labels and 2) leveraging coarse-level hierarchies to create auxiliary coarse labels. By incorporating these two lightweight auxiliary tasks, the proposed method can improve defect classification and facilitate defect detection. Furthermore, we collect and release a fabric defect dataset from two factories, containing 24 annotated categories. Each sample is captured under three different illumination conditions: back, front, and side lighting. Extensive experiments on public datasets, including TILDA, ZJU-Leaper, Tianchi, MVTec-AD, and our dataset, validate the effectiveness of the proposed method, which consistently achieves improvements of over 1% in both the classification accuracy and mean average precision (mAP)@0.5.**

***Index Terms*—Class hierarchy, deep learning, fabric dataset, fabric defect classification, fabric defect detection.**

I. Introduction

FABRIC defects are flaws or irregularities on textile surfaces, some of which can significantly affect product quality and price [1]. Defect detection is a vital aspect of quality control in the textile industry, enabling both localization and classification of defects. Currently, defect inspection in production mainly relies on manual labor, which poses several limitations: 1) low inspection speed reduces production efficiency; 2) operator fatigue often leads to missed or inconsistent detection results; and 3) when category-specific defect handling is required, accuracy is limited due to the large variety of defect categories. Consequently, there is an urgent demand in modern textile manufacturing for automatic fabric defect detection methods based on machine vision [2].

Fabric defect detection can be divided into traditional machine vision methods [1] and deep learning-based methods [2], [3], [4]. Most traditional methods are based on feature engineering with prior domain knowledge, including statistical [5], spectral-based [6], and template-matching methods [7]. Some classical machine learning methods, such as clustering [8], dictionary learning [9], [10], and support vector machines [11], are also incorporated in defect detection. Recently, deep learning has progressed rapidly and been applied in many fields [12]. Many studies have applied deep learning to fabric defect detection [13], [14], [15], [16] or standalone defect classification [17], [18], [19], [20]. In unsupervised settings, autoencoder-based methods [21] are often employed to estimate anomaly regions. Furthermore, with labeled samples, convolutional neural networks (CNNs) are trained to classify image patches [22] or whole images [23], [24]. When detailed annotations, such as bounding boxes, are available, advanced object detection methods can be transferred into defect detection. Two-stage detection methods [25], [26], known for their higher accuracy, operate in two stages: 1) in the first stage, a region proposal network is trained to generate a set of candidate regions; and 2) in the second stage, the candidate regions are passed through a classifier and regressor to classify objects and refine the bounding boxes. To meet real-time detection requirements, single-stage [27] and anchor-free [28], [29] detection methods are widely adopted in industrial settings, albeit with a tradeoff in performance. Considering the characteristics of fabric images, specific modules, such as SoftPool [30], atrous convolution [13], [34], and attention mechanisms [31], [32], [33], [34], have been designed or integrated to improve existing classifiers or detectors.

The aforementioned methods are primarily designed for binary classification tasks, i.e., defect versus nondefect. Although some methods support multiclass classification, they are often applied to binary or small-scale multiclass problems due to the lack of multiclass datasets. However, real-world fabric images may contain dozens of defect types [35], [36]. Since fabric defects vary in severity, factories often annotate and classify defects at different levels of granularity according to their specific needs, enabling class-specific handling. Therefore, multiclass classification is important for fabric defect detection [20], [37]. Existing research mainly addresses

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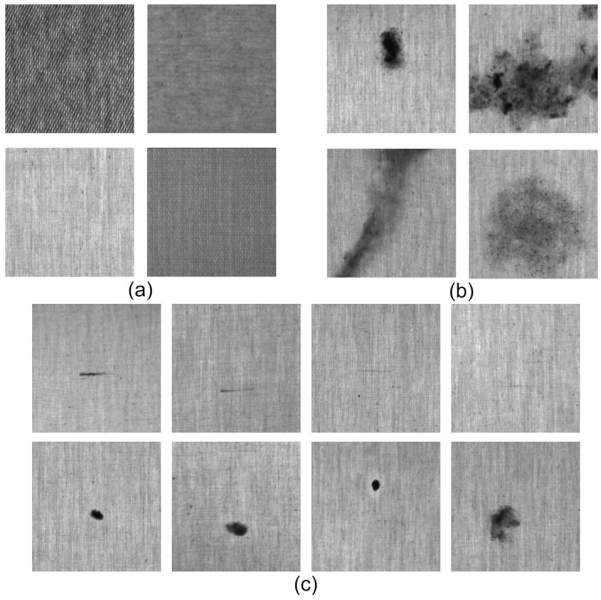


Fig. 1. Samples illustrating the category hierarchy problem in fabric defect classification. (a) Fabric images containing multiple textures. (b) Single annotated category (Category 23) with images belonging to various distinct subclasses. (c) Visual similarity among different annotated categories (top: Categories 13, 14, 16, and 17; bottom: Categories 20–23).

challenges such as class imbalance [38], [39] and data scarcity [23], [41]. In this study, we present two key problems in fabric defect classification: 1) some annotated defect categories contain multiple distinct subcategories but lack manual fine-grained annotations; and 2) annotated defect categories exhibit a hierarchical structure, which is not leveraged in training. For the first problem, as shown in Fig. 1(a) and (b), samples may contain various textile patterns, and a defect category may include several subcategories. This problem is particularly evident in binary classification, where all defects are grouped into a single class. For the second problem, as shown in Fig. 1(c), we can generate coarser labels by grouping the annotated categories based on visual similarity. Neglecting these two problems may limit the classification performance.

A large amount of research in computer vision has focused on flat classification problems [42], [43]. In those problems, each sample is assigned a single ground truth label, and all other classes are treated as mutually exclusive and equally distinct. However, in real-world applications, such as ImageNet [44], categorical labels are often organized hierarchically. Across public natural image datasets, studies [45], [46] have demonstrated that leveraging hierarchical information can lead to improved performance compared to flat classification. Moreover, when manually annotated category labels are overly coarse-grained, as in binary classification, the resulting hierarchical structure is very simple. Existing studies focus on adopting unsupervised learning [47], [48] or coarse labels [49], [50] to derive fine-grained or subcategory labels. However, little attention has been given to automatically mining subcategory information to improve the deep classifier for the annotated categories.

In recent years, several fabric defect detection or classification datasets [5], [51], [53], [54] have been made publicly available. However, in most of these datasets, defect categories are coarsely defined, and the fabric images are captured under a single lighting condition. When fine-grained classification is required, relying on a single lighting condition makes it difficult to distinguish between similar defect categories. Therefore, we collect and publicly release a new dataset in which samples are annotated into 24 categories. Each sample consists of three grayscale images captured under front, back, and side illumination. In this study, we create coarse labels and generate subcategory pseudo-labels. However, the main goal is to improve the classification or detection performance on the annotated defect categories, thereby better meeting practical requirements. We conduct extensive experiments on both our collected dataset and public benchmarks, covering fabric defect classification and detection tasks. The results demonstrate the effectiveness of the proposed method and highlight the importance of hierarchical information in fabric defect classification. In summary, our contributions are given as follows.

1) We propose that mining and leveraging category hierarchy information can improve the performance of deep classifiers and detectors in fabric inspection. Creating coarse labels from the annotated categories is straightforward. Moreover, subcategory mining is particularly valuable, as fabric images exhibit diverse patterns, and a single defect category may contain multiple distinct subclasses.
2) To facilitate mining subcategories and leveraging coarse labels, we employ a novel classification head compatible with any backbone network. This head supports flexible definitions of category hierarchies. Furthermore, we develop corresponding loss functions to enable effective multitask learning.
3) We release a real-world, multiclass fabric defect dataset to support research on fine-grained classification. The dataset includes 24 annotated categories. Specifically, to further enhance interclass visual discriminability among defect categories, we provide images captured under three different illumination conditions.

II. Related Works

A. Deep Learning Fabric Defect Detection and Classification

Deep learning has been widely applied to fabric defect detection and classification in both academic research and industrial applications. Unsupervised methods [55], [56], [57], [58] can detect defects without labeled data. A common strategy involves using a deep convolutional autoencoder [21], [57] for background reconstruction. Subtracting the reconstruction from the original image, a residual map can highlight potential regions. To enhance pixelwise detection, Tian and Li [58] proposed a novel autoencoder-based detection method utilizing cross-patch similarity. Hu et al. [55] extended the deep convolutional generative adversarial network (DCGAN) to reconstruct the query image. In addition, knowledge distillation [59], [60] has also been adopted in unsupervised anomaly detection. In the training process, a student network is supervised by the outputs of a fixed teacher network on normal images. In the test process, for anomaly images, the teacher

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and the student will output significantly different features, which can be used to compute a score map to locate anomalies.

Based on labeled samples, advanced deep learning classification and detection networks have been directly applied to fabric defect inspection [61], [62]. Jing et al. [22] trained CNNs to classify local patches and used a sliding window over the whole image during inference. Kahraman and Durmuşoğlu [17] used capsule networks to further improve the performance. Zahra et al. [63] adopted a saliency-based region technique to localize the defective areas and a simple CNN to classify the fabric images into defective and nondefective. Object detection networks were widely employed in fabric defect detection, including one-stage methods such as the YOLO series [27] and two-stage methods such as the R-CNN family [25], [64]. Wei et al. [37] used feature pyramid networks to detect multiclass fabric defects. Revathy and Kalaivani [65] further applied Mask R-CNN to the HKU dataset [51]. Other two-stage frameworks like Faster R-CNN [66] and Cascade R-CNN [67] have also been adapted for fabric defect detection tasks. YOLO models treat object detection as a single regression problem, enabling real-time inference. Various versions, including YOLOv5 [15], [68], YOLOv8 [69], and YOLOX [70], have been extensively applied to fabric defect detection.

Instead of using vanilla detection methods, novel neural networks have been specifically designed based on the characteristics of fabric defect detection. Zhao et al. [31] introduced a novel attention-based multiscale feature fusion module to enhance defect identification. Xiang et al. [13] proposed a hook-shaped feature pyramid network (HookFPN) to improve detection efficiency while maintaining accuracy. Cui et al. [71] developed the SDDNet to tackle large texture variations and subtle defects. Wang et al. [34] proposed a feature contrast interference suppression method that integrates contrastive learning, attention mechanisms, and knowledge distillation to address the challenges posed by background perturbation. An and Wan [39] introduced a joint dynamic context and balanced learning framework to mitigate the long-tail distribution problem in fabric defect detection. In addition to visual images, tactile images have also been employed to assist detection [32], [72].

B. Hierarchical Classification

Hierarchical label structures widely exist in many real-world datasets, such as ImageNet [44] and CIFAR [73]. Generally, leveraging class hierarchy is effective in improving deep classifiers [45], [46]. Some studies [74] introduced branches in CNNs to support hierarchical classification, often trained using multitask learning paradigms. Label-embedding methods [75], [76] map class labels to vectors whose relative locations represent class relationships. Several hierarchical loss functions have also been proposed, such as quad loss [77] and hierarchical cross-entropy (HXE) [45]. In addition, Zeng et al. [78] proposed cophenetic correlation coefficient regularization to embed class hierarchy in learned representations. Compared to classical flat classification [42], [43], incorporating hierarchical structure information can enable classifiers to make better mistakes and effectively improve fine-level classification accuracy. Furthermore, fine-grained visual classification [79] aims to discriminate closely related subclasses, e.g., different species of birds or plants. However, in real-world applications, some categories are insufficiently annotated, and additional labels can be prohibitively expensive. Deep clustering [47], [48] performs automatic classification based solely on image features. Self-training [80], pseudo-labeling [81], or mutual information maximization [82] are generally adopted to implement deep clustering. In addition, several studies have explored the use of coarse labels to guide automatic fine-grained classification [49] or feature representation learning [50].

Existing hierarchical classification methods mainly organize annotated labels into a hierarchical structure but do not provide additional supervision during training. Obtaining fine-grained subcategory labels, however, requires substantial manual effort. In this work, we aim to improve the performance on annotated categories by providing additional supervision through generating pseudo-subcategory labels. A straightforward approach is to automatically generate subcategory labels for each annotated class using techniques such as deep clustering [47] and then perform hierarchical classification by combining these subcategory labels with the annotated labels. However, this procedure is typically conducted in a stepwise manner. To overcome this limitation, we propose a unified framework that jointly optimizes clustering and supervised learning, guiding the clustering process to improve classification performance.

C. Fabric Datasets

Public annotated datasets are important in developing and benchmarking AI algorithms. In the field of fabric defect classification and detection, an increasing number of datasets have been publicly released. Textile texture database (TILDA) [5], which contains 3200 images, was released in 1996. The HKU-Fabric dataset [51], whose samples are generated and annotated based on simulation, was released by the University of Hong Kong in 2009. In 2019, a fabric dataset with bounding box annotations and multiclass defect labels was released by Alibaba Cloud Tianchi Platform [52]. Zhang et al. [53] published a large dataset named ZJU-Leaper in 2020. The dataset contains 98 777 samples with annotations for defect segmentation masks. The samples are categorized as defect or nondefect, with a portion of them collected from real factories. Carrilho et al. [54] released the Lusitano dataset, which consists of 36 000 images from a real factory. This dataset is primarily designed for evaluating one-class classification algorithms, as the training dataset only contains normal images. The images in these datasets are captured under a single lighting condition, and the defects are typically not finely categorized. To address these limitations, this study introduces a new dataset collected from two textile factories. It contains images captured under three different lighting conditions and covers 23 types of fabric defects, enabling more comprehensive evaluation and development of defect classification models.

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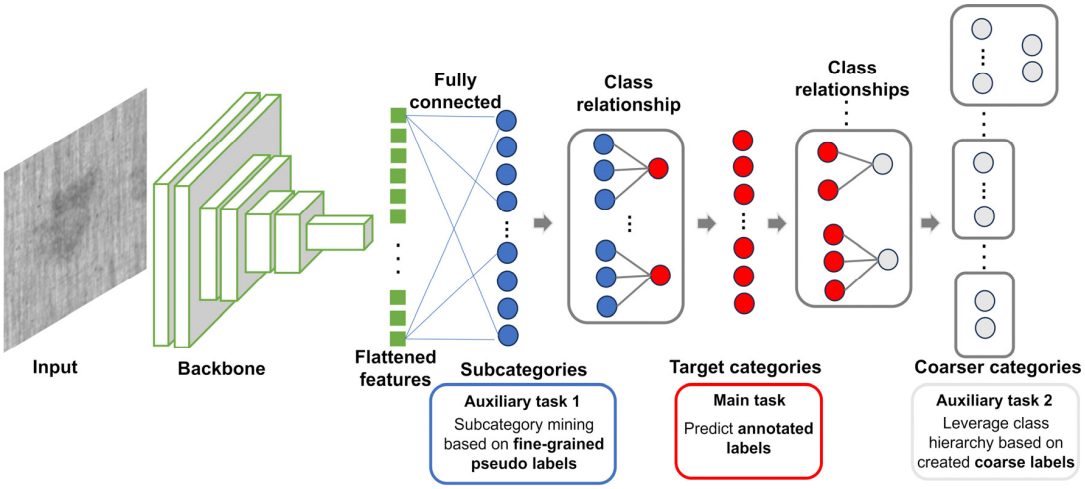


Fig. 2. Structure of the proposed method.

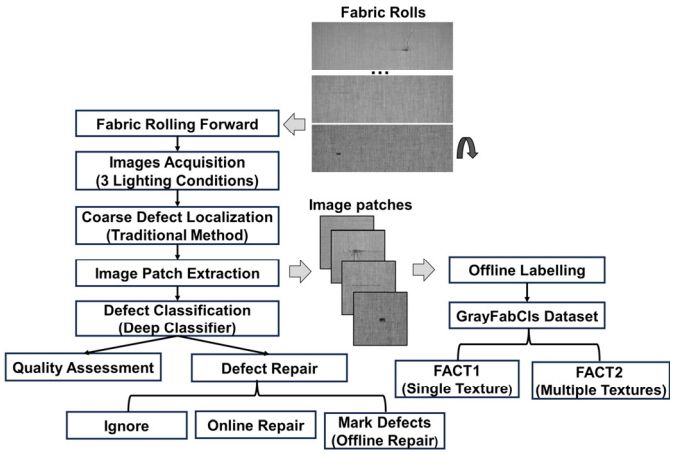


Fig. 3. Workflow of online defect detection and sample collection for the GrayFabCls dataset.

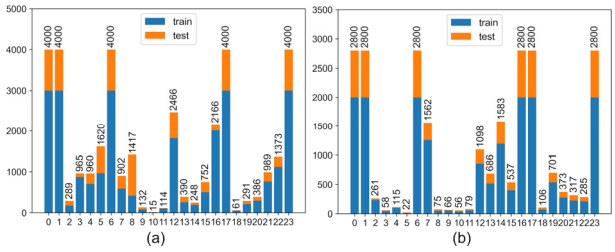


Fig. 4. Data distribution in the GrayFabCls dataset. (a) Fact1. (b) Fact2.

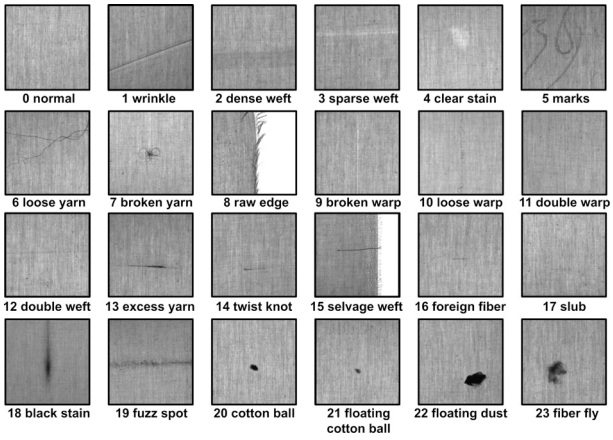


Fig. 5. Representative samples from each category in the GrayFabCls dataset.

III. Methods

As illustrated in Fig. 2, our proposed method is primarily designed for the classification task. However, since existing object detection methods typically contain a classification head, our approach can be seamlessly integrated into detection networks. In the experimental section, we demonstrate that our method not only improves classification performance but also enhances the mean average precision (mAP) in defect detection tasks. Details of our released dataset are provided in Section III-A. In Section III-B, we introduce the overall framework and problem setup. Section III-C presents the auxiliary task of subcategory mining in a supervised manner. Section III-D describes how we leverage the class hierarchy by defining category relationships and creating coarse labels. Finally, in Section III-E, we define the total loss function of our method and provide practical guidelines for hyperparameter tuning.

A. Dataset Overview

Fig. 3 illustrates the workflow of our online defect detection system and the sample collection procedure. As the fabric rolls forward, images under three different lighting conditions are captured by rapidly switching the light sources. Since highly precise localization is not required in actual production, a set of traditional algorithms is employed to perform coarse defect localization, and image patches are extracted for classification. In the early stage, these patches are manually annotated to construct the GrayFabCls dataset, which is subsequently used to train a deep classifier. After deployment, the system is able to classify defects, thereby supporting fabric quality assessment and enabling category-specific handling strategies, such as ignoring negligible defects, performing online repair, or marking defects for offline repair.

The GrayFabCls dataset consists of image patches collected from two textile factories and is divided into two subsets: Fact1 and Fact2. The fabric images in Fact1 contain only a single dominant texture, whereas those in Fact2 exhibit a variety of

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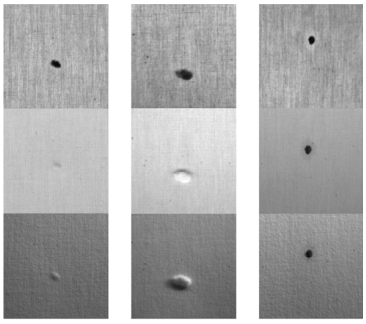


Fig. 6. Samples from Categories (a) 20, (b) 21, and (c) 22 in the GrayFabCls dataset, collected as three grayscale images under three different illumination conditions.

textures. The image patches in Fact1 have a resolution of 256×240, while those in Fact2 are 256×196. The image patches are categorized into 24 classes, including one defect-free class. Fig. 4 shows the statistical information of the GrayFabCls dataset, and Fig. 5 presents representative samples under back illumination. In fact, each sample includes three grayscale image patches captured under three illumination conditions: back, front, and side lighting. Under a single light source, distinguishing between certain defects is challenging due to their similar visual patterns. As shown in Fig. 6, Categories 20–22 are difficult to differentiate using only the back-illuminated images. Considering images captured under all three lighting conditions significantly improves their discriminability. During training, the three image patches are fed into the model as three separate channels.

As shown in Fig. 1(a), the same defect appearing on different background textures will be treated as the same category. Moreover, Fig. 1(b) highlights that certain categories, such as Category 23, comprise several visually distinguishable but unannotated subcategories. In contrast, as shown in Fig. 1(c), some defect categories (e.g., Categories 13, 14, 16, and 17) exhibit high visual similarity. Our method takes these two characteristics into account and leverages them to improve the performance of the deep classifier.

B. Framework Description

In fabric defect detection, the input is typically a whole image or an image patch, and the output consists of the bounding boxes of defect objects along with their class probability distributions. In this section, we focus on the classification problem and describe our proposed method in detail. Let $Y_{t\arg et}$ denote the set of annotated categories or target categories, which may vary according to different scenarios and factories. A labeled dataset $D = \{(\mathbf{x}^i, y^i_{t\arg et})\}_{i=1}^N$ is collected, where $\mathbf{x}^i$ is the input and $y^i_{t\arg et} \in Y_{t\arg et}$ is the annotated class label. In detection tasks, bounding box annotations are also provided. Our goal is to train a model that can predict bounding boxes and their categories from Y_{target} for a given test image.

In this study, two auxiliary tasks are introduced to improve the deep classifiers and indirectly enhance the defect detectors. First, a single defect category in Y_{target} may encompass several visually distinct subcategories, especially in binary classification settings. Even when fine-grained categories are manually annotated, the background fabric texture often exhibits diverse patterns that can themselves be further divided into additional subcategories. However, annotating subcategories is highly time-consuming and often lacks practical significance. Nevertheless, we hypothesize that subcategory mining could improve the classification performance on the target category set $Y_{t\arg et}$. Let Y_{sub} denote all subcategories. Second, $Y_{t\arg et}$ may exhibit a hierarchical structure, or a hierarchy can be constructed by treating the original classes in $Y_{t\arg et}$ as leaf nodes and grouping them based on visual similarity or production rules. Through these constructed hierarchical structures, we can easily obtain coarse labels for each sample, which can then also be used to improve the classification performance on $Y_{t\arg et}$. Let Y_{coarse} denote all coarse categories.

C. Subcategory Mining

As shown in Fig. 2, a fully connected layer, followed by a softmax function, is applied to the flattened backbone features to compute the probability distribution $\mathbf{p}_{sub}$ over Y_{sub}. To obtain target predictions $\mathbf{p}_{t\arg et}$ over $Y_{t\arg et}$, we define a mapping function that establishes the class relationship between Y_{sub} and $Y_{t\arg et}$. The mapping function is shown as follows:

$$\mathbf{p}_{t\arg et} = \mathbf{M}_{sub}\mathbf{p}_{sub} \tag{1}$$

where $\mathbf{M}_{sub}$ is a binary matrix. $\mathbf{M}^{i,j}_{sub}$ is 1 if the jth subcategory in Y_{sub} is associated with the ith category in $Y_{t\arg et}$. In the experiment section, we introduce a global hyperparameter, *numsub*, which specifies the number of subcategories into which each target category is divided. To facilitate stable and effective training, we employ two regularizations on $\mathbf{p}_{sub}$ to guide the learning process.

For each sample, we aim to concentrate the probability distribution on a single subcategory. Following the approach used in deep clustering [83], we construct a target distribution $\mathbf{q}_{sub}$ as a pseudo-label, which is shown as follows:

$$\mathbf{q}^{ij}_{sub} = \frac{\mathbf{q}^2_{ij} \Big/ \mathbf{f}_j}{\sum_{j'} \mathbf{q}^2_{ij'} / \mathbf{f}_{j'}} \tag{2}$$

$$\mathbf{q}_{ij} = \begin{cases} \mathbf{p}^{ij}_{sub}, & if\ \mathbf{M}^{l_i,j}_{sub} = 1 \\ 0, & otherwise \end{cases} \tag{3}$$

where $\mathbf{p}^{ij}_{sub}$ denotes the predicted probability of assigning sample i to subcategory j. l_i represents the class label of the ith sample in Y_{target}. $\mathbf{f}_j = \sum_i \mathbf{p}^{ij}_{sub}$ is a soft frequency, which is used to mitigate the problem of class imbalance. The corresponding regularization term is to minimize the cross-entropy between $\mathbf{q}_{sub}$ and $\mathbf{p}_{sub}$

$$R_{sharp} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} \mathbf{q}^{ij}_{sub} \log \mathbf{p}^{ij}_{sub}. \tag{4}$$

To prevent degenerate solutions, where samples are classified into a single subcategory, we introduce a regularization term by minimizing the Kullback–Leibler (KL) divergence between the mean $\mathbf{p}_{sub}$ and a prior probability distribution $\mathbf{p}_{sub-prior}$. The prior $\mathbf{p}_{sub-prior}$ assumes a uniform distribution

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TABLE I
CREATED COARSE CATEGORIES BASED ON VISUAL SIMILARITY AND PRODUCTION RULE FOR THE GRAYFABCLS DATASET

Rules	Coarse category	Annotated category
Visual similarity	0	0,1,2,3,4
	1	5,6,7,8
	2	9,10,11
	3	12,13,14,15,16,17
	4	18,19,20,21,22,23
Production rule: defect vs. non-defect	0	0,1,6,17,21,22,23
	1	2,3,4,5,7,8,9,10,11,12 13,14,15,16,18,19,20

over the subcategories within each category, and $\mathbf{M}_{sub}\mathbf{p}_{sub-prior}$ is aligned with the overall distribution of the training samples. This regularization term [84] is shown as follows:

$$R_{uniform} = \sum_{c=1}^{C} \mathbf{p}^{c}_{sub-prior} \log\left(\frac{\mathbf{p}^{c}_{sub-prior}}{\widehat{\mathbf{p}}^{c}_{sub}}\right) \tag{5}$$

where $\widehat{\mathbf{p}}^{c}_{sub}$ denotes the mean probability of the model for class c across all samples in one mini-batch. C is the number of subcategories in Y_{sub}.

D. Leverage Coarse Labels

The defect class set Y_{target} is typically defined according to business-specific requirements. However, the categories in Y_{target} can be grouped or structured based on certain rules, forming coarse-level labels. For instance, in the GrayFabCls dataset, the 24 classes can be divided into five groups according to the visual similarity of the defects, as shown in Table I. In practical production, some defects require a machine shutdown for repair, while others do not. Based on this rule, the 24 classes can be divided into two groups, also listed in Table I. As shown in Fig. 2, after obtaining the predicted class probabilities $\mathbf{p}_{target}$ over Y_{target}, we can define a class relationship between Y_{target} and Y_{coarse}. Given one such relationship, the mapping function is shown as follows:

$$\mathbf{p}_{coarse} = \mathbf{M}_{coarse}\mathbf{p}_{target} \tag{6}$$

where $\mathbf{M}_{coarse}$ is a binary matrix that encodes the predefined class relationship. $\mathbf{M}^{i,j}_{coarse}$ is 1 if the jth category in Y_{target} is associated with the ith category in Y_{coarse}.

E. Loss Function and Hyperparameter Tuning

Considering the regularizations for subcategory mining, annotated labels, and created coarse labels, we define the total loss for our proposed deep classifier as follows:

$$\begin{aligned} Loss = {} & focal\left(\mathbf{p}_{target}, y_{target}\right) + \lambda_a R_{sharp} + \lambda_b R_{uniform} \\ & + \lambda_c focal\left(\mathbf{p}_{coarse}, y_{coarse}\right) \end{aligned} \tag{7}$$

where $focal$ denotes the focal loss [85], which is commonly used to mitigate class imbalance. λ_a, λ_b, and λ_c are hyperparameters that balance the contribution of each loss term. Although three hyperparameters are introduced, the individual loss terms are of similar magnitude. Below, we present guidelines for tuning these hyperparameters.

1) For λ_a, it should be chosen so that the value of R_{sharp} is approximately on the same scale as the focal loss $focal(\mathbf{p}_{target}, y_{target})$ during the early stages of training.
2) λ_b controls the strength of subcategory mining and can be tuned within the range of 0.0–10.0.
3) If only one type of coarse label is created, λ_c can be simply set to 1.0. When multiple types of coarse labels are created, λ_c decreases inversely with respect to the number of coarse label types.

IV. EXPERIMENTS AND DISCUSSION

In this section, we evaluate the effectiveness of our proposed method on five datasets: TILDA [51], ZJU-Leaper [53], Tianchi [52], MVTec-AD [86], and GrayFabCls. We first briefly introduce each dataset and the experimental settings in this study. Second, we summarize the evaluation metrics. Finally, we present comparative experiments and ablation studies to demonstrate the effectiveness of our method.

A. Datasets and Experimental Settings

TILDA contains a total of eight representative textile types and defines seven defect classes. Although some defective images are accompanied by defect descriptions in German, corresponding bounding boxes are not available. In our study, we only use the data in the cd1 folder (subfolders: c1r1, c1r3, c2r2, and c2r3), which contains four textile types. The dataset is randomly split into 80% for training and 20% for testing. Classification performance is evaluated. ZJU-Leaper contains 19 fabric types, but the defect categories are not explicitly labeled. Images collected in the laboratory tend to yield artificially high performance. Therefore, in this study, we only focus on images obtained from real-world factories. These cover four fabric types and contain 4862 defective images. The Tianchi dataset contains 5913 defective images with a high resolution of 2440 × 1000 pixels. The defect categories are finely annotated with 34 classes. However, some categories were officially merged, resulting in 20 classes that are used for performance evaluation. Since bounding box annotations are available, we evaluate the detection performance on the ZJU-Leaper and Tianchi datasets. To further assess the generality of our method across domains, we evaluate its detection performance on the MVTec-AD dataset, which contains 15 categories of industrial products. Since MVTec-AD is originally designed for unsupervised defect detection, we reorganize its annotated test set of 1725 samples by splitting it into training and testing subsets. GrayFabCls is our newly collected dataset, which is described in Section III-A. In this study, we separately evaluate classification performance on its two subsets: Fact1 and Fact2.

For those five datasets, all experiments are conducted on a workstation with the Ubuntu Linux operating system. The GPU is an NVIDIA GeForce RTX 4090. We initialize our models using pretrained weights from OpenMMLab's [87]

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open-source model repository. The classification model is fine-tuned for ten epochs in total, including a one-epoch warm-up phase. The AdamW optimizer [88], with a weight decay of 0.0001, is used. The base learning rate is set to 0.001, and a cosine decay [89] strategy is employed for learning rate scheduling. Class imbalance is addressed by applying inverse class frequency weighting. Only minimal augmentation, such as random horizontal flipping, is used. In our experiments, ConvNeXt-tiny [90] is adopted as the baseline model for classification tasks, while YOLOv5-small [27] serves as the baseline for detection tasks. The detection model is fine-tuned for 30 epochs, including a three-epoch warm-up phase. For the baseline model, only the manually annotated labels are used for supervision. To improve performance on the annotated categories, our method introduces two auxiliary tasks: subcategory mining and leveraging coarse labels. Specifically, in the model with auxiliary task 1, fine-grained pseudo-labels are automatically mined. In the model with auxiliary task 2, coarse labels are constructed as auxiliary supervision by hierarchically organizing the annotated categories. In addition, based on the GrayFabCls dataset, we compare and integrate our method with the category hierarchy-based approach in [45], which employs the ResNet-18 [42] backbone and incorporates HXE and soft-label techniques. The hyperparameters are set to $\alpha = 0.1$ for HXE and $\beta = 15$ for the soft-label method. Each experiment is repeated five times with different random seeds, and the average results are reported.

B. Evaluation Metrics

For classification experiments, we use two commonly adopted evaluation metrics: accuracy and macro-F1. Accuracy is defined as follows:

$$Accuracy = \frac{N_{correct}}{N_{all}} \tag{8}$$

where $N_{correct}$ is the number of correctly classified samples and N_{all} is the total number of test samples. F1 is defined by the following formulas:

$$P = \frac{TP}{TP + FP} \tag{9}$$

$$R = \frac{TP}{TP + FN} \tag{10}$$

$$F1 = \frac{2 \times P \times R}{P + R} \tag{11}$$

where TP, FP, and FN are the numbers of true positives, false positives, and false negatives, respectively. To obtain macro-F1, we first calculate $F1$ for each category and then take the mean value over all categories.

For detection experiments, we use mAP to evaluate detection performance. The average precision (AP) is a standard metric used to evaluate object detection performance for a single class. It is calculated as the area under the precision–recall curve. The mAP is the average AP value across all classes. In textile factories, the intersection over union (IoU) threshold for bounding boxes is generally set to a relatively low value. In this study, we report mAP under the IoU threshold of 0.5 (mAP@0.5) unless otherwise specified.

TABLE II

CLASSIFICATION RESULTS (%) ON THE TILDA DATASET FOR DIFFERENT SETTINGS. ACC@8 AND ACC@2 DENOTE THE EIGHT-CLASS AND BINARY CLASSIFICATION METRICS, RESPECTIVELY

Settings	Acc@8	macro-F1 @8	Acc@2	macro-F1 @2
ConvNeXt-tiny@8	94.38	94.38	98.88	97.48
w/ aux task 1	95.56	95.54	99.13	98.06
w/ aux task 2	95.25	95.23	**99.19**	**98.19**
w/ aux tasks 1&2	**95.63**	**95.64**	98.94	97.64

For datasets with imbalanced categories, we also compare the classwise AP or $F1$-score and report the number of classes that achieve performance improvements among those exhibiting differences greater than 1%.

C. Results on the Defect Classification

For the TILDA dataset, ConvNeXt-tiny is used as the baseline model. We incorporate auxiliary task 1 (subcategory mining), auxiliary task 2 (leveraging coarse labels), and both tasks. In auxiliary task 1, since the dataset contains four textile types, we set $numsub = 4$. We also set $\lambda_a = 1.0$ and $\lambda_b = 1.0$. In auxiliary task 2, we create binary coarse labels based on defect/nondefect grouping. The original images, with a resolution of 768×512, are resized to 256×256 before being fed into the network. In the training process, we train an eight-class classification model. In the testing phase, we evaluate performance on both the eight-class classification task and the binary classification task (defect versus nondefect). The experimental results are summarized in Table II. By incorporating either one or both auxiliary tasks, the accuracy and macro-F1-score of the eight-class classification model can be improved by approximately 1%. The best performance is achieved when both auxiliary tasks are jointly utilized. Subcategory mining yields greater performance improvements than leveraging coarse labels on this dataset. In terms of binary classification metrics, leveraging coarse labels alone results in the largest improvement. Although subcategory mining also improves binary classification metrics, jointly utilizing those two tasks leads to a slight performance drop.

D. Results on the Defect Detection

For the detection task, YOLOv5-small is adopted as the baseline. In the ZJU-Leaper dataset, 3252 samples are used for training, and 1610 samples are used for performance evaluation. We categorize the defects into four classes based on fabric type. Given the small number of annotated categories, we do not create coarse labels and only incorporate auxiliary task 1. We set $numsub = 4$, $\lambda_a = 0.2$, and $\lambda_b = 5.0$. Images are resized to 256×256. Experimental results are reported in Table III. The baseline YOLOv5-small model achieves an mAP of 66.12%. After incorporating the auxiliary task of subcategory mining, the model achieves an improvement of 1.2%.

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TABLE III
DETECTION RESULTS (%) ON THE ZJU-LEAPER, MVTEC-AD, AND TIANCHI DATASETS FOR DIFFERENT SETTINGS

Dataset	Settings	mAP@0.5
ZJU-Leaper	YOLOv5-small@4	66.12
	w/ aux task 1	**67.38**
MVTec AD	YOLOv5-small@15	67.04
	w/ aux task 1	**69.10**
Tianchi	YOLOv5-small@20	27.84
	w/ aux task 1	29.96
	w/ aux task 2	28.00
	w/ aux tasks 1&2	**31.36**

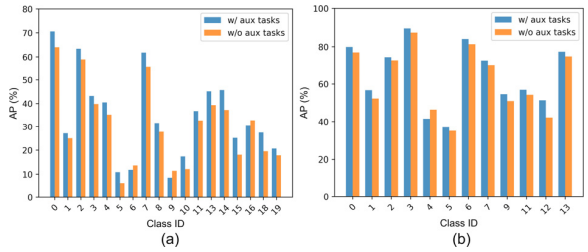


Fig. 7. Comparison of classwise AP for categories with differences exceeding 1%. (a) Tianchi dataset. (b) MVTec-AD dataset.

In the Tianchi dataset, all defective images are randomly split into 70% for training and 30% for testing. Images are resized to 1216 × 480. We set *numsub* = 4, λ_a = 0.5, λ_b = 7.0, and λ_c = 1.0. The experimental results are reported in Table III. The baseline model achieves an mAP of 27.84%. Since some categories are merged into a single class and the fabric types are diverse, incorporating the auxiliary task of subcategory mining can significantly improve the detection performance (27.84% versus 29.96%). Regarding auxiliary task 2, the 20 categories are divided into four groups to form coarse labels. Specially, these four groups are: 1) light-colored small defects (Categories 0, 2, 3, 11, 12, and 16); 2) dark-colored small defects (Categories 1, 6, and 14); 3) warp-direction long defects (Categories 7–10, 15, 18, and 19); and 4) weft-direction long defects (Categories 4, 5, 13, and 17). The mAP is improved by 0.16% after incorporating auxiliary task 2. Finally, incorporating both auxiliary tasks achieves the best result, with the mAP improved by 3.5%. In addition, as shown in Fig. 7(a), we also compare the classwise AP with and without incorporating the auxiliary tasks, showing that 15 out of the 18 categories achieve performance improvements.

In the MVTec-AD dataset, all samples from the official test dataset are randomly split into 70% for training and 30% for testing. All images are resized to 512 × 512 pixels. We train a 15-class defect detection model. Since the annotated categories are mutually independent, only auxiliary task 1 is incorporated. We set *numsub* = 5, λ_a = 0.5, and λ_b = 5.0. The experimental results are summarized in Table III. The baseline YOLOv5-small model achieves an mAP of 67.04%, while incorporating the subcategory mining task brings a 2.06% improvement. Furthermore, Fig. 7(b) presents a comparison of the AP across categories, showing that 11 out of the 12 categories exhibit performance improvements.

TABLE IV
HYPERPARAMETERS USED FOR EACH MODEL AND DATA SUBSET

Factory	Model	λ_a	λ_b	λ_c
Fact1	Binary	0.05	3.0	/
	24-class	0.5	3.0	1.0
Fact2	Binary	0.05	7.5	/
	24-class	0.5	5.0	1.0

TABLE V
CLASSIFICATION RESULTS (%) ON THE FACT1 SUBSET FOR DIFFERENT SETTINGS

Settings	Acc@24	macro-F1 @24	Acc@2	macro-F1 @2
ConvNeXt-tiny@2	/	/	93.26	93.07
w/ aux task 1	/	/	**93.75**	**93.57**
ConvNeXt-tiny@24	82.80	73.78	94.06	93.91
w/ aux task 1	**83.76**	74.02	94.04	93.85
w/ aux task 2	83.02	74.04	93.93	93.76
w/ aux tasks 1&2	83.74	**74.54**	**94.46**	**94.30**

E. Results on the GrayFabCls Dataset

For our GrayFabCls dataset, we adopt ConvNeXt-tiny as the baseline model. Given the annotations for 24 categories, both auxiliary tasks can be incorporated. The dataset contains images from two factories: Fact1 and Fact2. We train and evaluate models on each subset separately. For auxiliary task 1, we set *numsub* = 5, even though Fact2 includes more fabric types. For auxiliary task 2, we construct binary classification labels (defect versus nondefect) based on the production rules in those two factories. In practical scenarios, some types of defects, although technically anomalous, have minimal impact on the final product and are therefore treated as nondefective. In this dataset, classes {0, 1, 6, 17, 21, 22, and 23} are considered nondefective, while the remaining classes are regarded as defective. In the testing phase, the 24-class classification model is used to evaluate both the 24-class performance and the two-class performance. In addition, using only the binary labels, we also train a binary classification model and incorporate auxiliary task 1. Table IV summarizes the hyperparameter configurations used in these experiments.

For the Fact1 dataset, the experimental results are shown in Table V. This subset contains one main fabric pattern. However, some defect classes exhibit intraclass variability and can be further subdivided. For the binary classification model (ConvNeXt@2), incorporating subcategory mining leads to a 0.5% improvement in the evaluation metrics. For the 24-class model (ConvNeXt@24), mining subcategories and leveraging coarse labels both improve classification performance. When

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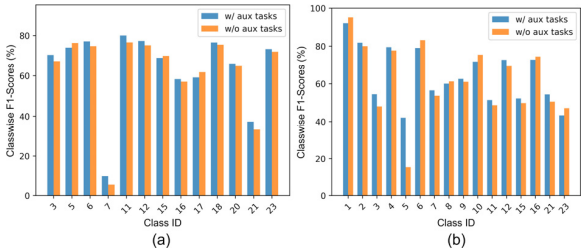


Fig. 8. Comparison of classwise $F1$-scores for categories with differences exceeding 1%. (a) Fact1 dataset. (b) Fact2 dataset.

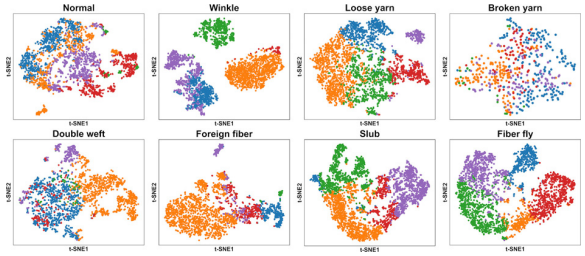


Fig. 9. t-SNE visualization of the learned features for the five subcategories of each annotated category in the Fact1 dataset.

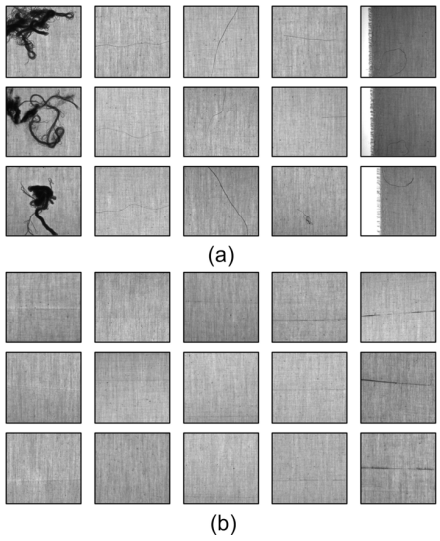


Fig. 10. Top-3 representative samples for the five subcategories within each annotated category of the Fact1 dataset. (a) Loose yarn. (b) Double weft.

both auxiliary tasks are incorporated, the model achieves a 1% improvement in accuracy and a 0.8% improvement in macro-F1. Notably, for binary classification, directly training with fine-grained labels (ConvNeXt@24) outperforms training with only binary labels (ConvNeXt@2). This result supports our hypothesis that subcategory information can improve the deep classifier for the annotated categories. Nevertheless, the automatic mining of subcategories still underperforms compared to directly training with manually annotated subcategory labels (93.75% versus 94.06%). The latter approach requires more human effort. Due to the class imbalance, we also compare the classwise $F1$-scores with and without the auxiliary tasks. As shown in Fig. 8(a), 10 out of 13 categories achieve performance improvements.

TABLE VI
CLASSIFICATION RESULTS (%) ON THE FACT1 SUBSET COMPARED WITH THE METHOD IN [45]

Settings	Acc@24	macro-F1 @24	Avg. hier. dist. @1	Hier. dist. mistake
CE	79.75	70.35	0.5876	2.918
w/ aux task 1	79.66	71.27	0.5838	2.891
w/ aux tasks 1&2	79.65	71.29	0.5855	2.896
HXE ($\alpha = 0.1$)	79.52	69.03	0.6020	2.957
w/ aux task 1	79.09	69.71	0.5972	2.873
soft-labels ($\beta = 15$)	80.87	70.99	0.5445	2.863
w/ aux task 1	**80.88**	**71.50**	**0.5357**	**2.821**

Although our primary objective is to improve the classification performance on the annotated categories, we further demonstrate the effectiveness of subcategory mining by visualizing the learned features for several categories using t-SNE [91], as shown in Fig. 9. Different subcategories are represented by distinct colors. For the loose yarn and double weft defects, we present the top-3 highest probability samples from each of the five subcategories, as shown in Fig. 10. The results indicate that the automatically mined subcategories exhibit clear feature differences. Thus, our method not only eliminates the need for manual fine-grained labeling but also effectively exploits the mined subcategory information.

Furthermore, we compare our method with the approach proposed in [45], which leverages class hierarchies to make better mistakes in image classification. Since the method in [45] does not involve subcategory mining, it can be naturally integrated with our proposed auxiliary task 1. In addition to accuracy and macro-F1-score, we also evaluate the hierarchical distance of misclassifications (hier. dist. mistake) and the average hierarchical distance of the top-1 predictions (avg. hier. dist.@1), where the hierarchy is consistent with that used in auxiliary task 2. The experimental results are summarized in Table VI. For the baseline, we adopt ResNet-18 trained with the standard cross-entropy (CE) loss. Incorporating the auxiliary tasks leads to improvements in macro-F1 and reductions in hierarchical distance, though the overall accuracy shows a slight decrease. The HXE-based method underperforms both the baseline and our proposed method; when combined with auxiliary task 1, it achieves a higher macro-F1-score but a lower overall accuracy. In contrast, the soft-label method consistently outperforms the baseline across all metrics, and its integration with auxiliary task 1 achieves the best overall performance.

For the Fact2 dataset, the experimental results are summarized in Table VII. This subset includes multiple fabric types. Subcategory mining alone yields an improvement of 1.1% in classification accuracy (78.78% versus 79.88%). Both auxiliary tasks positively contribute to performance. Incorporating both auxiliary tasks leads to the best results. For the 24-class classification model (ConvNeXt@24), the accuracy and macro-F1-score are improved by 1.2% and 1.5%, respectively. ConvNeXt@24 outperforms ConvNeXt@2

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TABLE VII

CLASSIFICATION RESULTS (%) ON THE FACT2 SUBSET FOR DIFFERENT SETTINGS

Settings	Acc@24	macro-F1 @24	Acc@2	macro-F1 @2
ConvNeXt-tiny@2	/	/	88.89	88.30
w/ aux task 1	/	/	**89.36**	**88.75**
ConvNeXt-tiny@24	78.78	66.95	89.38	88.84
w/ aux task 1	79.88	67.57	**90.22**	89.62
w/ aux task 2	79.29	66.59	89.95	89.35
w/ aux tasks 1&2	**80.05**	**68.47**	90.16	**89.56**

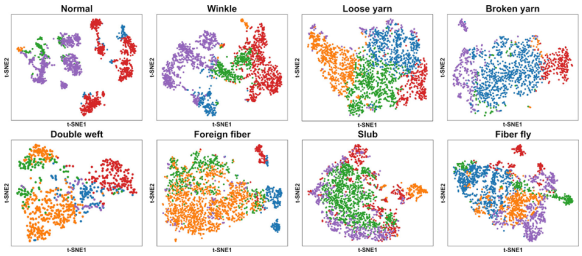


Fig. 11. t-SNE visualization of the learned features for the five subcategories of each annotated category in the Fact2 dataset.

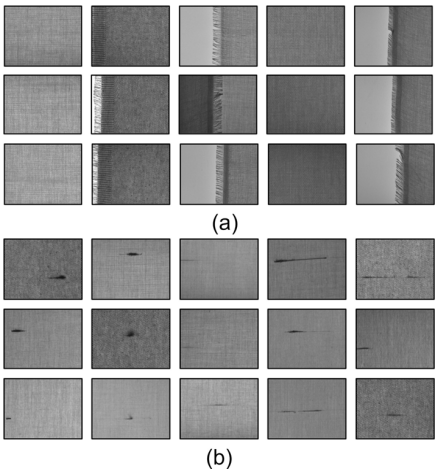


Fig. 12. Top-3 representative samples for the five subcategories within each annotated category of the Fact2 dataset. (a) Normal. (b) Excess yarn.

in binary classification metrics. This further highlights the importance of subcategory information. Notably, the performance of ConvNeXt@2 with auxiliary task 1 approaches that of ConvNeXt@24, which is directly supervised with manually annotated subcategory labels (89.36% versus 89.38%). Therefore, in real-world production scenarios, even when binary classification is the primary concern, it remains beneficial to manually annotate fine-grained labels or incorporate our subcategory mining method. Fig. 8(b) presents a comparison of classwise $F1$-scores after incorporating the auxiliary tasks, showing that 11 out of the 16 categories exhibit performance improvements. Regarding the effectiveness of subcategory mining, Fig. 11 presents t-SNE visualizations of the learned features for several categories, highlighting the distinct feature differences between subcategories. For the normal samples,

TABLE VIII

CLASSIFICATION RESULTS (%) ON THE FACT2 SUBSET COMPARED WITH THE METHOD IN [45]

Settings	Acc@24	macro-F1 @24	Avg. hier. dist. @1	Hier. dist. mistake
CE	77.63	62.63	0.6784	3.031
w/ aux task 1	78.67	**65.78**	**0.6414**	3.005
w/ aux tasks 1&2	**78.72**	64.78	0.6438	3.023
HXE ($\alpha = 0.1$)	77.47	63.36	0.6817	3.024
w/ aux task 1	78.33	65.51	0.6505	**3.000**
soft-labels ($\beta = 15$)	77.64	62.65	0.6735	3.009
w/ aux task 1	78.22	64.34	0.6552	3.006

Fig. 12(a) displays the top-3 highest probability samples from each subcategory, revealing clear texture differences among subcategories. Fig. 12(b) illustrates representative samples for each subcategory of the excess yarn defect. It can be observed that, for defect samples, subcategories are primarily divided based on defect morphology rather than background texture. When deep clustering is applied alone, background texture tends to dominate the clustering process. By jointly training classification and clustering, our method guides the clustering process to learn representations that are more discriminative for defect classification.

Furthermore, we evaluate our method on the Fact2 dataset by comparing and integrating it with the HXE and soft-label approaches. The experimental results are summarized in Table VIII. Except for HXE alone, all evaluated combinations outperform the baseline. Incorporating auxiliary task 1 alone achieves the highest macro-F1-score and the lowest average hierarchical distance, whereas combining both auxiliary tasks 1 and 2 achieves the highest classification accuracy. Moreover, when starting from either HXE or the soft-label method, further integrating auxiliary task 1 consistently improves classification performance and reduces hierarchical distance, resulting in macro-F1 gains exceeding 1.5%.

F. Ablative Study

In this section, we conduct ablation studies mainly on the Fact2 subset and the Tianchi dataset to investigate the impact of key modules and hyperparameters on the effectiveness of our proposed method.

Subcategory Mining: To support subcategory mining, our method introduces two regularization terms, R_{sharp} and $R_{uniform}$, along with three corresponding hyperparameters, λ_a, λ_b, and *numsub*. λ_a controls the concentration of the subcategory probability distribution, while λ_b controls the strength of automatic subcategory mining. Fig. 13(b) shows the accuracy curve with respect to λ_b. When $\lambda_b = 0$, only R_{sharp} takes effect, and it also outperforms the baseline model. When $\lambda_b > 0$, $R_{uniform}$ further helps to improve the classification accuracy. When $\lambda_b > 4.0$, the model performance becomes relatively insensitive to λ_b. Fig. 13(a) shows the accuracy curve with respect to λ_a. After incorporating R_{sharp}, the model consistently outperforms the baseline model in

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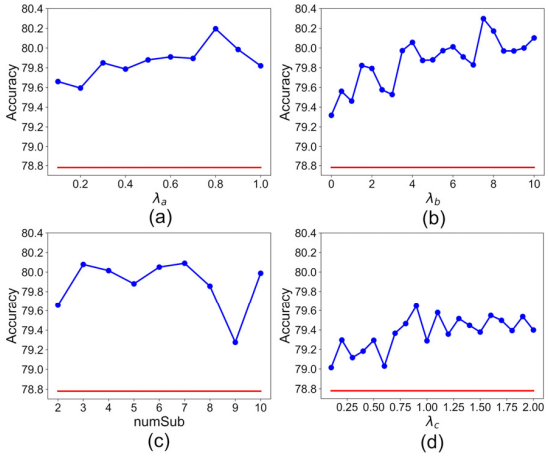


Fig. 13. Accuracy curves on the Fact2 test set versus (a) λ_a, (b) λ_b, (c) *numsub*, and (d) λ_c. The baseline model accuracy is 78.78%.

TABLE IX
CLASSIFICATION RESULTS (%) ON THE FACT2 SUBSET BASED ON DIFFERENT COARSE LABEL SETTINGS

Settings	Acc@24	macro-F1 @24
ConvNeXt-tiny@24	78.78	66.95
w/ binary	79.29	66.59
w/ 5-class	78.87	67.14
w/ binary&5-class	**79.63**	**67.79**

terms of accuracy. Fig. 13(c) shows the accuracy curve with respect to *numsub*, which determines how many subcategories each original category is divided into. As *numsub* increases, the model performance initially improves but then begins to decline. The optimal *numsub* depends on the number of fabric types within the dataset and the granularity of the annotated defect categories. Specifically, we may set *numsub* for each category.

Leveraging Coarse Labels: In the above experiments, coarse labels are created solely based on a binary rule (defect versus nondefect). Fig. 13(d) shows the accuracy curve with respect to λ_c. Leveraging coarse labels consistently improves the model performance on the annotated categories. The accuracy increases as λ_c increases, and then decreases afterward. In addition, other types of coarse labels can be created, or multiple coarse label types can be leveraged simultaneously. In the GrayFabCls dataset, the annotated categories can also be grouped by visual similarity, forming five-class coarse labels as shown in Table I. Table IX summarizes the performance of models under different coarse label settings. With $\lambda_c = 1.0$, leveraging only the five-class coarse labels slightly improves accuracy and macro-F1. Leveraging only binary coarse labels improves accuracy but leads to a drop in the macro-F1. When both types of coarse labels are leveraged simultaneously, we set $\lambda_c = 0.5$. Both accuracy and macro-F1 achieve the best performance. Therefore, constructing additional coarse labels has the potential to further enhance the deep classifiers.

Given the importance of subcategory mining, we further conduct a sensitivity study on the hyperparameter *numsub* in

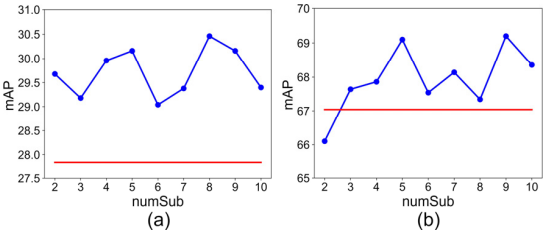


Fig. 14. mAP curves versus *numsub* on (a) Tianchi dataset and (b) MVTec-AD dataset. The baseline mAP values for (a) and (b) 27.84 and 67.04, respectively.

TABLE X
CLASSIFICATION RESULTS (%) ON THE FACT2 SUBSET FOR DIFFERENT BACKBONES

Settings	Acc@24	macro-F1 @24
ResNet-18	77.61	**65.36**
w/ aux tasks 1&2	**78.34**	64.57
EfficientNet-b0	76.08	62.00
w/ aux tasks 1&2	**76.62**	**62.74**
ViT-base	71.23	60.84
w/ aux tasks 1&2	**74.17**	**61.70**
Swin-base	74.06	61.25
w/ aux tasks 1&2	**76.25**	**62.32**
ShuffleNetV2	74.68	61.16
w/ aux tasks 1&2	**76.23**	**62.23**
MobileViT-xxsmall	75.01	62.66
w/ aux tasks 1&2	**76.81**	**63.11**

the object detection task. Fig. 14(a) illustrates the variation of mAP with respect to *numsub* on the Tianchi dataset. With the other hyperparameters fixed, incorporating subcategory mining consistently yields a clear performance improvement over the baseline. However, the magnitude of this improvement does not show a strong dependence on the specific value of *numsub*. Fig. 14(b) presents the variation of mAP with respect to *numsub* on the MVTec-AD dataset. Except for the case where *numsub* = 2, which results in a lower mAP than the baseline, all other settings outperform the baseline. Moreover, no strong correlation is observed between *numsub* and the degree of performance improvement. Nevertheless, setting *numsub* = 5 achieves comparatively good results. Notably, most categories in the MVTec-AD dataset [86] consist of five defect groups, which provides practical guidance for choosing this value.

Different Backbones: In the previous experiments, we only adopted ConvNeXt as the backbone. To investigate the effectiveness of our method across different architectures, we conduct experiments on several widely used models, covering both standard and lightweight networks. These include ResNet [42], EfficientNet [92], Vision Transformers [43], Swin Transformers [93], ShuffleNet [94], and MobileViT [95]. All experiments use the same hyperparameters, differing only in the backbone architectures. The experimental results are summarized in Table X. Our method improves performance across all backbones except for a drop in macro-F1 observed

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TABLE XI
DETECTION RESULTS (%) ON THE TIANCHI DATASET FOR YOLOV8 AND YOLOV11 UNDER DIFFERENT SETTINGS

Settings	mAP@0.5
YOLOv8n	29.96
w/ aux task 1	**30.97**
w/ aux task 2	30.63
w/ aux tasks 1&2	30.91
YOLOv11n	33.89
w/ aux task 1	**35.12**
w/ aux task 2	34.14
w/ aux tasks 1&2	34.62

with ResNet-18. Notably, our method achieves greater performance gains on Transformer-based models, with accuracy improvements reaching up to 3%. Furthermore, as shown in Fig. 3, under the available computational resources, all evaluated models satisfy the production requirements for fabric roll speed, and the additional overhead introduced by our method is negligible.

For the object detection task, we have already demonstrated the performance improvements achieved by integrating our method with YOLOv5. Based on the Tianchi dataset, we further incorporate our method into more advanced real-time object detection models from the Ultralytics framework [96], including YOLOv8 and YOLOv11. Both YOLOv8 and YOLOv11 adopt task-aligned learning [97] in their classification heads, where the classification loss is formulated as multiple binary CE (BCE) losses rather than a multiclass CE loss. Consequently, the probability relationship between parent categories and subcategories must be redefined. Specifically, if a parent category contains k subcategories, its probability is computed as

$$p_{parent} = 1 - \prod_{i=1}^{k} (1 - p_i) \tag{12}$$

where p_{parent} denotes the probability of the parent category and p_i denotes the probability of the ith subcategory. Experimental results in Table XI show that, for both YOLOv8 and YOLOv11, integrating either auxiliary task 1 or auxiliary task 2 individually, as well as jointly integrating both auxiliary tasks, consistently improves detection performance. Notably, incorporating the subcategory mining task alone yields the largest performance gain for both models, leading to an improvement of over 1% in mAP.

Robustness: In the above experiments, we report the average performance over five runs with different seeds to mitigate performance fluctuations due to randomness. In this section, we further analyze the robustness of the performance improvements achieved by our method. We first increase the number of random seeds to 10 per experiment and report additional statistical measures. Fig. 15(a) shows the accuracy variations across different random seeds for all models on the Fact2 dataset, while Fig. 15(b) shows the corresponding mAP variations

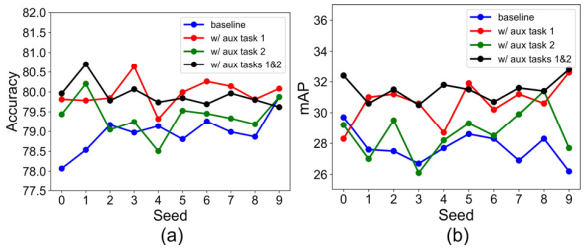


Fig. 15. Accuracy and mAP under different settings over ten independent runs with different random seeds: (a) accuracy on the Fact2 dataset and (b) mAP on the Tianchi dataset.

TABLE XII
CLASSIFICATION AND DETECTION PERFORMANCE OVER TEN RUNS (MEAN±STD), INCLUDING PAIRED T-TEST T-STATISTICS AND 95% CONFIDENCE INTERVALS COMPARED TO BASELINE

Settings	$mean \pm std$ (Acc or mAP)	paired t-test t-stat	95% CI (Δ)
ConvNeXt-tiny@24	78.87 ± 0.37	/	/
w/ aux task 1	79.96 ± 0.37	6.85	[0.72,1.46]
w/ aux task 2	79.33 ± 0.45	1.94	[−0.09,1.00]
w/ aux tasks 1&2	$\mathbf{79.95 \pm 0.31}$	5.56	[0.63,1.53]
YOLOv5-small@20	27.92 ± 0.92	/	/
w/ aux task 1	30.41 ± 1.19	4.15	[1.10, 3.87]
w/ aux task 2	28.79 ± 1.58	1.75	[−0.27,2.01]
w/ aux tasks 1&2	$\mathbf{31.33 \pm 0.62}$	13.37	[2.82,4.00]

TABLE XIII
CLASSIFICATION AND DETECTION PERFORMANCE ON ILLUMINATION-TRANSFORMED TEST DATASET OVER TEN RUNS (MEAN±STD), INCLUDING PAIRED T-TEST T-STATISTICS AND 5% CONFIDENCE INTERVALS COMPARED TO BASELINE

Settings	$mean \pm std$ (Acc or mAP)	paired t-test t-stat	95% CI (Δ)
ConvNeXt-tiny@24	78.64 ± 0.32	/	/
w/ aux task 1	79.34 ± 0.57	3.01	[0.17,1.24]
w/ aux task 2	79.10 ± 0.56	1.82	[−0.12,1.05]
w/ aux tasks 1&2	$\mathbf{79.47 \pm 0.30}$	4.44	[0.40,1.26]
YOLOv5-small@20	26.21 ± 0.80	/	/
w/ aux task 1	28.62 ± 0.81	5.33	[1.37, 3.45]
w/ aux task 2	27.29 ± 1.15	2.62	[0.13,2.03]
w/ aux tasks 1&2	$\mathbf{29.52 \pm 0.77}$	14.79	[2.79,3.83]

on the Tianchi dataset. Table XII summarizes the statistics over ten10 runs. Across both datasets, incorporating both auxiliary tasks yields the highest average performance and the lowest variance. Across the 10 runs, performance differences (accuracy or mAP) between our method and the baseline are computed and evaluated using two-sided paired t-tests. The resulting t-statistics are large, and the 95% confidence intervals for the mean differences are entirely positive, indicating that the observed improvements are statistically consistent and robust.

We further apply random illumination transformations to the test dataset and evaluate all models to assess robustness to illumination changes. The statistical results are summarized

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in Table XIII. Since only random flipping is used for data augmentation during training, the models exhibit some performance degradation on the illumination-transformed dataset. However, the performance gains achieved by our method remain stable, with minimal changes in the 95% confidence intervals. In addition, the Fact2 dataset with illumination transformations will also be made publicly available.

Finally, we also evaluate the stability of the clustering process in subcategory mining. Pseudo-subcategory labels for each sample across the ten runs on the Fact2 dataset are predicted. Clustering stability is quantified using normalized mutual information (NMI) across all 45 possible pairs of runs. On the training set, clustering shows high consistency, with an average NMI of 0.7334 and a variance of 0.0056. The test set exhibits slightly lower stability, with an average NMI of 0.6786 and a variance of 0.0075.

V. Conclusion

Fabric defect categories are often annotated at varying levels of granularity, ranging from binary to fine-grained multiclass classifications, depending on practical requirements. A deep classifier or detector is then trained accordingly. However, in annotated datasets, each defect category can typically be further subdivided, as multiple fabric types or distinct defect patterns may be grouped under a single label. Moreover, the annotated category labels can be structured hierarchically to create coarser labels. In this study, we aim to investigate this hierarchical information to improve the deep classifier for the annotated categories. Our proposed method comprises two main components: mining subcategories and leveraging coarse labels. The class hierarchy is encoded using a nonlearnable network. We employ R_{sharp} to concentrate the probability distribution over subcategories, which serves as pseudo-labels. $R_{uniform}$ encourages each annotated category to be subdivided. Furthermore, the classifier can be supervised with coarse labels, which are generated by grouping the annotated class labels based on visual similarity or production rules. We also introduce GrayFabCls, a real-world fabric dataset with fine-grained category annotations and images captured under three illumination conditions. Extensive experiments are conducted on the TILDA, ZJU-Leaper, Tianchi, and GrayFabCls datasets. Our method consistently improves classifier performance with minimal additional computation overhead and also contributes to improving the detector. We also conduct ablation studies on key modules and network backbones. In the future, we plan to apply our method to other industrial applications.

References

[1] A. Kumar, "Computer-vision-based fabric defect detection: A survey," *IEEE Trans. Ind. Electron.*, vol. 55, no. 1, pp. 348–363, Jan. 2008, doi: 10.1109/TIE.1930.896476.

[2] Y. Kahraman and A. Durmuşoğlu, "Deep learning-based fabric defect detection: A review," *Textile Res. J.*, vol. 93, nos. 5–6, pp. 1485–1503, Mar. 2023, doi: 10.1177/00405175221130773.

[3] C. Li, J. Li, Y. Li, L. He, X. Fu, and J. Chen, "Fabric defect detection in textile manufacturing: A survey of the state of the art," *Secur. Commun. Netw.*, vol. 2021, May 2021, Art. no. 9948808, doi: 10.1155/2021/9948808.

[4] R. Khanam, M. Hussain, R. Hill, and P. Allen, "A comprehensive review of convolutional neural networks for defect detection in industrial applications," *IEEE Access*, vol. 12, pp. 94250–94295, 2024, doi: 10.1109/ACCESS.2024.3425166.

[5] D. Chetverikov and A. Hanbury, "Finding defects in texture using regularity and local orientation," *Pattern Recognit.*, vol. 35, no. 10, pp. 2165–2180, Oct. 2002, doi: 10.1016/s0031-3203(01)00188-1.

[6] P. Li, H. Zhang, J. Jing, R. Li, and J. Zhao, "Fabric defect detection based on multi-scale wavelet transform and Gaussian mixture model method," *J. Textile Inst.*, vol. 106, no. 6, pp. 587–592, Jun. 2015, doi: 10.1080/00405000.2014.929790.

[7] H. Y. T. Ngan, G. K. H. Pang, S. P. Yung, and M. K. Ng, "Wavelet based methods on patterned fabric defect detection," *Pattern Recognit.*, vol. 38, no. 4, pp. 559–576, Apr. 2005, doi: 10.1016/j.patcog.2004.07.009.

[8] J. G. Campbell, C. Fraley, F. Murtagh, and A. E. Raftery, "Linear flaw detection in woven textiles using model-based clustering," *Pattern Recognit. Lett.*, vol. 18, no. 14, pp. 1539–1548, Dec. 1997, doi: 10.1016/s0167-8655(97)00148-7.

[9] X. Kang and E. Zhang, "A universal and adaptive fabric defect detection algorithm based on sparse dictionary learning," *IEEE Access*, vol. 8, pp. 221808–221830, 2020, doi: 10.1109/ACCESS.2020.3041849.

[10] P. Li, J. Liang, X. Shen, M. Zhao, and L. Sui, "Textile fabric defect detection based on low-rank representation," *Multimedia Tools Appl.*, vol. 78, no. 1, pp. 99–124, Jan. 2019, doi: 10.1007/s11042-017-5263-z.

[11] A. Kumar and H. C. Shen, "Texture inspection for defects using neural networks and support vector machines," in *Proc. Int. Conf. Image Process.*, vol. 1, Sep. 2002, pp. III-353–III-356, doi: 10.1109/icip.2002.1038978.

[12] M. H. M. Noor and A. O. Ige, "A survey on state-of-the-art deep learning applications and challenges," 2025, *arXiv:2403.1756*.

[13] Z. Xiang, Y. Shen, M. Ma, and M. Qian, "HookNet: Efficient multiscale context aggregation for high-accuracy detection of fabric defects," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023, doi: 10.1109/TIM.2023.3280519.

[14] L. Shao, E. Zhang, Q. Ma, and M. Li, "Pixel-wise semisupervised fabric defect detection method combined with multitask mean teacher," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022, doi: 10.1109/TIM.2022.3162286.

[15] Y. Guo, X. Kang, J. Li, and Y. Yang, "Automatic fabric defect detection method using AC-YOLOv5," *Electronics*, vol. 12, no. 13, p. 2950, Jul. 2023, doi: 10.3390/electronics12132950.

[16] Z. Jia, Z. Shi, Z. Quan, and M. Shunqi, "Fabric defect detection based on transfer learning and improved faster R-CNN," *J. Eng. Fibers Fabrics*, vol. 17, pp. 1–18, Jan. 2022, doi: 10.1177/15589250221086647.

[17] Y. Kahraman and A. Durmuşoğlu, "Classification of defective fabrics using capsule networks," *Appl. Sci.*, vol. 12, no. 10, p. 5285, May 2022, doi: 10.3390/app12105285.

[18] Z. Zhan, J. Zhou, and B. Xu, "Fabric defect classification using prototypical network of few-shot learning algorithm," *Comput. Ind.*, vol. 138, Jun. 2022, Art. no. 103628, doi: 10.1016/j.compind.2022.103628.

[19] F. G. Yaşar Çıklaçandır, S. Utku, and H. Özdemir, "Determination of various fabric defects using different machine learning techniques," *J. Textile Inst.*, vol. 115, no. 5, pp. 733–743, May 2024, doi: 10.1080/00405000.2023.2201978.

[20] T. Meeradevi and S. Sasikala, "Automatic fabric defect detection in textile images using a labview based multiclass classification approach," *Multimedia Tools Appl.*, vol. 83, no. 25, pp. 65753–65772, Jan. 2024, doi: 10.1007/s11042-023-18087-7.

[21] C. Lv, Z. Zhang, F. Shen, F. Zhang, and H. Su, "A fast surface defect detection method based on background reconstruction," *Int. J. Precis. Eng. Manuf.*, vol. 21, no. 3, pp. 363–375, Mar. 2020, doi: 10.1007/s12541-019-00262-2.

[22] J.-F. Jing, H. Ma, and H.-H. Zhang, "Automatic fabric defect detection using a deep convolutional neural network," *Coloration Technol.*, vol. 135, no. 3, pp. 213–223, Jun. 2019, doi: 10.1111/cote.12394.

[23] B. Lu, M. Zhang, and B. Huang, "Deep adversarial data augmentation for fabric defect classification with scarce defect data," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022, doi: 10.1109/TIM.2022.3185609.

[24] T. Almeida, F. Moutinho, and J. P. Matos-Carvalho, "Fabric defect detection with deep learning and false negative reduction," *IEEE Access*, vol. 9, pp. 81936–81945, 2021, doi: 10.1109/ACCESS.2021.3086028.

[25] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017, doi: 10.1109/TPAMI.2016.2577031.

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[26] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[27] J. Terven, D.-M. Córdova-Esparza, and J.-A. Romero-González, "A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS," *Mach. Learn. Knowl. Extraction*, vol. 5, no. 4, pp. 1680–1716, Nov. 2023, doi: 10.3390/make5040083.

[28] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[29] X. Wang, W. Fang, and S. Xiang, "Fabric defect detection based on anchor-free network," *Meas. Sci. Technol.*, vol. 34, no. 12, Aug. 2023, Art. no. 125402, doi: 10.1088/1361-6501/ace8af.

[30] Q. Liu, C. Wang, Y. Li, M. Gao, and J. Li, "A fabric defect detection method based on deep learning," *IEEE Access*, vol. 10, pp. 4284–4296, 2022, doi: 10.1109/ACCESS.2021.3140118.

[31] Y. Zhao et al., "Attention-based multiscale feature fusion for efficient surface defect detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–10, 2024, doi: 10.1109/TIM.2024.3372229.

[32] B. Fang, X. Long, F. Sun, H. Liu, S. Zhang, and C. Fang, "Tactile-based fabric defect detection using convolutional neural network with attention mechanism," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–9, 2022, doi: 10.1109/TIM.2022.3165254.

[33] J. Tang, H. Zhang, Z. Su, M. Jiang, P. Li, and J. Jing, "PUNet: Printed fabric defect segmentation method based on UNet," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025, doi: 10.1109/TIM.2025.3550640.

[34] J. Wang, J. Cheng, C. Gao, J. Zhou, and L. Shen, "Enhanced fabric defect detection with feature contrast interference suppression," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025, doi: 10.1109/TIM.2025.3553976.

[35] S. Shahrabadi, Y. Castilla, M. Guevara, L. G. Magalhães, D. Gonzalez, and T. Adão, "Defect detection in the textile industry using image-based machine learning methods: A brief review," *J. Phys., Conf. Ser.*, vol. 2224, no. 1, Apr. 2022, Art. no. 012010, doi: 10.1088/1742-6596/2224/1/012010.

[36] Cotton Incorporated. *Standard Fabric Defect Glossary*. Accessed: Jun. 9, 2025. [Online]. Available: https://www.cottoninc.com/quality-products/textile-resources/fabric-defect-glossary/

[37] B. Wei, L. Gao, X.-S. Tang, and K. Hao, "Multi-class object learning with application to fabric defects detection," *AATCC J. Res.*, vol. 8, no. 1, pp. 165–172, Sep. 2021, doi: 10.14504/ajr.8.s1.20.

[38] J. Jing, Z. Wang, M. Rätsch, and H. Zhang, "Mobile-UNet: An efficient convolutional neural network for fabric defect detection," *Textile Res. J.*, vol. 92, nos. 1–2, pp. 30–42, Jan. 2022, doi: 10.1177/0040517520928604.

[39] N. An and L. Wan, "JDCBL: A joint dynamic contexts and balanced learning framework for real-world fabric defect detection," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025, doi: 10.1109/TIM.2025.3558828.

[40] J. Liu, C. Wang, H. Su, B. Du, and D. Tao, "Multistage GAN for fabric defect detection," *IEEE Trans. Image Process.*, vol. 29, pp. 3388–3400, 2020, doi: 10.1109/TIP.2019.2959741.

[41] C. Gao, X. Chen, J. Zhou, J. Wang, and L. Shen, "Open-set fabric defect detection with defect generation and transfer," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–13, 2025, doi: 10.1109/TIM.2025.3547485.

[42] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[43] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[44] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[45] L. Bertinetto, R. Mueller, K. Tertikas, S. Samangooei, and N. A. Lord, "Making better mistakes: Leveraging class hierarchies with deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12503–12512.

[46] D. Silva-Palacios, C. Ferri, and M. J. Ramírez-Quintana, "Improving performance of multiclass classification by inducing class hierarchies," *Proc. Comput. Sci.*, vol. 108, pp. 1692–1701, Jan. 2017, doi: 10.1016/j.procs.2017.05.218.

[47] Y. Ren et al., "Deep clustering: A comprehensive survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 5858–5878, Apr. 2025, doi: 10.1109/TNNLS.2024.3403155.

[48] S. Zhou et al., "A comprehensive survey on deep clustering: Taxonomy, challenges, and future directions," *ACM Comput. Surv.*, vol. 57, no. 3, pp. 1–38, Nov. 2024, doi: 10.1145/3689036.

[49] M. Grcić, A. Gadetsky, and M. Brbić, "Fine-grained classes and how to find them," 2024, *arXiv:2406.11070*.

[50] H. Touvron, A. Sablayrolles, M. Douze, M. Cord, and H. Jégou, "Grafit: Learning fine-grained image representations with coarse labels," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 854–864.

[51] H. Y. T. Ngan and G. K. H. Pang, "Regularity analysis for patterned texture inspection," *IEEE Trans. Autom. Sci. Eng.*, vol. 6, no. 1, pp. 131–144, Jan. 2009, doi: 10.1109/TASE.2008.917140.

[52] (2019). *Fabric Dataset*. [Online]. Available: https://tianchi.aliyun.com/competition/entrance/231748/information

[53] C. Zhang, S. Feng, X. Wang, and Y. Wang, "ZJU-leaper: A benchmark dataset for fabric defect detection and a comparative study," *IEEE Trans. Artif. Intell.*, vol. 1, no. 3, pp. 219–232, Dec. 2020, doi: 10.1109/TAI.2021.3057027.

[54] R. Carrilho, K. A. Hambarde, and H. Proença, "A novel dataset for fabric defect detection: Bridging gaps in anomaly detection," *Appl. Sci.*, vol. 14, no. 12, p. 5298, Jun. 2024, doi: 10.3390/app14125298.

[55] G. Hu, J. Huang, Q. Wang, J. Li, Z. Xu, and X. Huang, "Unsupervised fabric defect detection based on a deep convolutional generative adversarial network," *Textile Res. J.*, vol. 90, nos. 3–4, pp. 247–270, Feb. 2020, doi: 10.1177/0040517519862880.

[56] C. Wei, J. Liang, H. Liu, Z. Hou, and Z. Huan, "Multi-stage unsupervised fabric defect detection based on DCGAN," *Vis. Comput.*, vol. 39, no. 12, pp. 6655–6671, Dec. 2023, doi: 10.1007/s00371-022-02754-1.

[57] Z. Yu, Y. Xu, Y. Wang, Y. Wang, and X. Sheng, "Unsupervised fabric defect detection based on multiscale image reconstruction and structural similarity assessment," *Coloration Technol.*, vol. 140, no. 6, pp. 827–842, Dec. 2024, doi: 10.1111/cote.12744.

[58] H. Tian and F. Li, "Autoencoder-based fabric defect detection with cross-patch similarity," in *Proc. 16th Int. Conf. Mach. Vis. Appl. (MVA)*, May 2019, pp. 1–6.

[59] M. Salehi, N. Sadjadi, S. Baselizadeh, M. H. Rohban, and H. R. Rabiee, "Multiresolution knowledge distillation for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14897–14907.

[60] G. Wang, S. Han, E. Ding, and D. Huang, "Student-teacher feature pyramid matching for anomaly detection," 2021, *arXiv:2103.04257*.

[61] A. Zahra, M. Amin, F. E. A. El-Samie, and M. Emam, "Efficient utilization of deep learning for the detection of fabric defects," *Neural Comput. Appl.*, vol. 36, no. 11, pp. 6037–6050, Apr. 2024, doi: 10.1007/s00521-023-09137-0.

[62] M. Nasim, R. Mumtaz, M. Ahmad, and A. Ali, "Fabric defect detection in real world manufacturing using deep learning," *Information*, vol. 15, no. 8, p. 476, Aug. 2024, doi: 10.3390/info15080476.

[63] A. Zahra, M. Amin, F. E. A. El-Samie, and M. Emam, "Fabric defect detection based on saliency map and keypoints," *J. Opt.*, vol. 52, no. 4, pp. 1750–1757, Dec. 2023, doi: 10.1007/s12596-022-00989-8.

[64] J. Wu et al., "Automatic fabric defect detection using a wide-and-light network," *Int. J. Speech Technol.*, vol. 51, no. 7, pp. 4945–4961, Jul. 2021, doi: 10.1007/s10489-020-02084-6.

[65] G. Revathy and R. Kalaivani, "Fabric defect detection and classification via deep learning-based improved mask RCNN," *Signal, Image Video Process.*, vol. 18, no. 3, pp. 2183–2193, Apr. 2024, doi: 10.1007/s11760-023-02884-6.

[66] M. Chen et al., "Improved faster R-CNN for fabric defect detection based on Gabor filter with genetic algorithm optimization," *Comput. Ind.*, vol. 134, Jan. 2022, Art. no. 103551, doi: 10.1016/j.compind.2021.103551.

[67] L. Li, Q. Li, Z. Liu, and L. Xue, "Effective fabric defect detection model for high-resolution images," *Appl. Sci.*, vol. 13, no. 18, p. 10500, Sep. 2023, doi: 10.3390/app131810500.

[68] F. Li, K. Xiao, Z. Hu, and G. Zhang, "Fabric defect detection algorithm based on improved YOLOv5," *Vis. Comput.*, vol. 40, no. 4, pp. 2309–2324, Apr. 2024, doi: 10.1007/s00371-023-02918-7.

[69] Y. Jin et al., "A real-time fabric defect detection method based on improved YOLOv8," *Appl. Sci.*, vol. 15, no. 6, p. 3228, doi: 10.3390/app15063228.

[70] M. Mao and M. Hong, "YOLO object detection for real-time fabric defect inspection in the textile industry: A review of YOLOv1 to YOLOv11," *Sensors*, vol. 25, no. 7, p. 2270, Apr. 2025, doi: 10.3390/s25072270.

[71] L. Cui, X. Jiang, M. Xu, W. Li, P. Lv, and B. Zhou, "SDDNet: A fast and accurate network for surface defect detection," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–13, 2021, doi: 10.1109/TIM.2021.3056744.

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[72] S. Li, W. Sun, Q. Liang, J. Sun, and C. Liu, "Multimodal fabric defect classification using channel switching and multiscale feature fusion," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025, doi: 10.1109/TIM.2024.3470964.
[73] Y. Zheng, H. Huang, and J. Chen, "Comparative analysis of various models for image classification on cifar-100 dataset," *J. Phys., Conf. Ser.*, vol. 2711, no. 1, Feb. 2024, Art. no. 012015, doi: 10.1088/1742-6596/2711/1/012015.
[74] J. Zhao, Y. Peng, and X. He, "Attribute hierarchy based multi-task learning for fine-grained image classification," *Neurocomputing*, vol. 395, pp. 150–159, Jun. 2020, doi: 10.1016/j.neucom.2018.02.109.
[75] S. Bengio, J. Weston, and D. Grangier, "Label embedding trees for large multi-class tasks," in *Proc. Neural Inform. Process. Syst.*, Dec. 2010, pp. 163–171.
[76] Q. Shi, M. Ye, Z. Zhang, and B. Du, "Learnable hierarchical label embedding and grouping for visual intention understanding," *IEEE Trans. Affect. Comput.*, vol. 14, no. 4, pp. 3218–3230, Oct. 2023, doi: 10.1109/TAFFC.2023.3247876.
[77] H. Proença, E. Yaghoubi, and P. Alirezazadeh, "A quadruplet loss for enforcing semantically coherent embeddings in multi-output classification problems," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 800–811, 2021, doi: 10.1109/TIFS.2020.3023304.
[78] S. Zeng, R. T. des Combes, and H. Zhao, "Learning structured representations by embedding class hierarchy," in *Proc. 11th Int. Conf. Learn. Represent.*, 2022, pp. 1–12.
[79] X.-S. Wei et al., "Fine-grained image analysis with deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 8927–8948, Dec. 2022, doi: 10.1109/TPAMI.2021.3126648.
[80] M.-R. Amini, V. Feofanov, L. Pauletto, L. Hadjadj, É. Devijver, and Y. Maximov, "Self-training: A survey," *Neurocomputing*, vol. 616, Feb. 2025, Art. no. 128904, doi: 10.1016/j.neucom.2024.128904.
[81] M. Caron, P. Bojanowski, A. Joulin, and M. Douze, "Deep clustering for unsupervised learning of visual features," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 132–149.
[82] Q. Ji, Y. Sun, Y. Hu, and B. Yin, "Variational deep embedding clustering by augmented mutual information maximization," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 2196–2202.
[83] J. Xie, R. Girshick, and A. Farhadi, "Unsupervised deep embedding for clustering analysis," in *Proc. 33rd Int. Conf. Mach. Learn.*, Jan. 2016, pp. 478–487.
[84] E. Arazo, D. Ortego, P. Albert, N. E. O'Connor, and K. McGuinness, "Pseudo-labeling and confirmation bias in deep semi-supervised learning," 2019, *arXiv:1908.02983*.
[85] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020, doi: 10.1109/TPAMI.2018.2858826.
[86] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9584–9592.
[87] K. Chen et al., "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.
[88] P. Zhou, X. Xie, Z. Lin, and S. Yan, "Towards understanding convergence and generalization of AdamW," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 9, pp. 6486–6493, Sep. 2024, doi: 10.1109/TPAMI.2024.3382294.
[89] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," 2016, *arXiv:1608.03983*.
[90] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11976–11986.
[91] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Nov. 2008.
[92] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.
[93] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[94] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," 2018, *arXiv:1807.11164*.
[95] S. Mehta and M. Rastegari, "MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer," 2021, *arXiv:2110.02178*.
[96] G. Jocher, A. Chaurasia, and J. Qiu. (2025). *YOLO by Ultralytics*. [Online]. Available: https://github.com/ultralytics/ultralytics
[97] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3490–3499.



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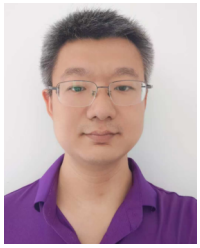
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